Section 1: Executive Summary

*I have completed a security review of the **Grand Marina Hydrologic System**. This includes all the smart technology we use to manage the marina, such as the digital water level sensors, automated flood pumps, and the remote-controlled valves that manage water flow. These systems are the "heart" of our daily operations, keeping the docks stable and the environment safe for our guests and their vessels.*

Threat modeling is essential because it allows us to find and fix "digital holes" before a real attacker finds them. Since our equipment is spread out across the water and connected to the internet, it is more vulnerable than a standard office computer. By identifying these risks now, we can prevent expensive equipment failures, environmental fines, or damage to our guests' boats that could hurt the marina's reputation and bottom line.

Through this analysis, I found three high-priority threats that need your attention:

1. ***System Blindness:** An attacker could trick our sensors into reporting "safe" water levels during a storm, which would stop our emergency pumps from turning on automatically.*
2. ***Unauthorized Access:** Currently, someone using our Guest Wi-Fi could potentially find a way into the controls for the main water valves.*
3. ***Lack of Backup Records:** It is currently too easy for someone to change system settings and delete the evidence, making it hard for us to know what went wrong if an accident happens.*

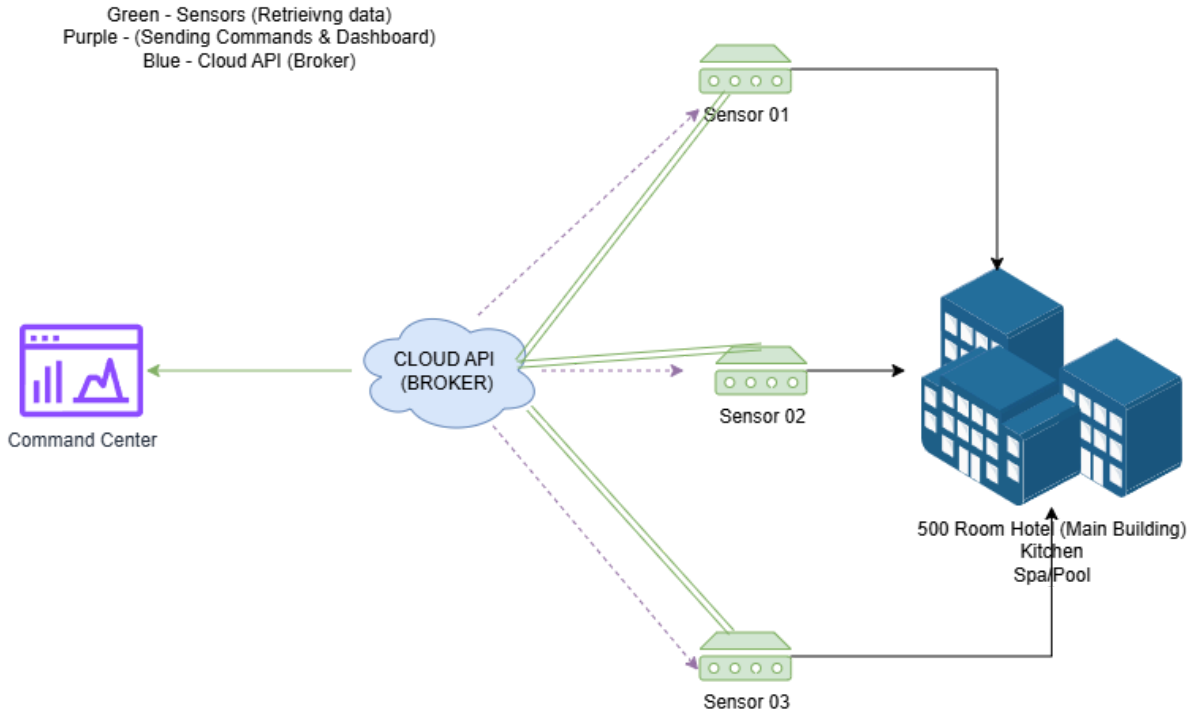
Section 2: System Overview

Describe The Grand Marina's HYDROLOGIC system:

- *500-room luxury hotel (Hydroficient customer)*
- *3 HYDROLOGIC flow management devices (one per water service line)*
- *Cloud-based monitoring and control*
- *Web dashboard for operators and management*
- *Remote shutoff and gate control capability*

Include a simple data flow diagram (can be ASCII art or describe the flow):

None



Section 3: Asset Inventory

List the key assets and their CIA priorities (use your work from Step 2):

Asset	Description	C Priority	I Priority	A Priority
HYDROLOGIC Devices	3 flow management units	Low	High	Critical
Web Dashboard	Operator monitoring interface	Medium	Critical	Critical
Cloud API	Device-to-cloud communication	Medium	Critical	Critical
Remote Controls	Gate adjustments, emergency shutoff	High	Critical	Critical
Consumption Data	Savings reports, billing records	Medium	High	Medium

Section 4: STRIDE Analysis

This is the core of your threat model. For **each major component**, analyze all six STRIDE categories:

Component 1: HYDROLOGIC Devices

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	Attacker impersonates a water level sensor to report "normal" levels during a flood surge, preventing automated pump activation.	Medium	High	High
Tampering	Physical or remote alteration of a water quality sensor's threshold settings, allowing toxic runoff to go undetected.	Medium	Medium	Medium
Repudiation	A contractor logs into the sluice gate controls and opens them incorrectly, but deletes system logs to deny they performed the action.	Low	Medium	Medium
Info Disclosure	Interception of unencrypted telemetry data revealing exact vessel coordinates or slip occupancy patterns to unauthorized third parties.	High	Medium	Medium
Denial of Service	Flooding the marina's wireless gateway with traffic, causing smart shut-off valves to become unresponsive during a pipe burst.	Medium	High	High
Elevation of Privilege	A guest uses the "Marina Wi-Fi" to exploit a vulnerability in a smart	Low	High	High

	dock controller, gaining administrative access to the entire hydrologic grid.			

Component 2: Web Dashboard

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	An attacker uses stolen session cookies (Session Hijacking) to masquerade as the Marina Manager and gain full dashboard access.	high	high	high
Tampering	An attacker uses an Insecure Direct Object Reference (IDOR) to change the "Safe Threshold" values for water pressure in the database.	Medium	High	Medium
Repudiation	The dashboard lacks a robust audit trail, allowing a disgruntled employee to disable alarm notifications without a record of the change.	Medium	Medium	Medium
Info Disclosure	Sensitive API keys or customer billing details are leaked through descriptive error messages or poorly configured cloud storage.	High	Medium	High
Denial of Service	A Distributed Denial of Service (DDoS) attack overwhelms the web server, leaving staff "blind" to real-time hydrologic alerts.	Medium	High	High
Elevation of Privilege	A standard "Viewer" account exploits a SQL	Low	High	High

	Injection vulnerability to gain "Super Admin" rights over the entire system.			

Component 3: Cloud API

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	An attacker uses a captured API Key to send fake "critical low water" signals from a rogue device to the cloud.	High	High	High
Tampering	A "Man-in-the-Middle" (MitM) attack modifies API JSON payloads in transit to change pump speed commands.	Medium	High	High
Repudiation	The API fails to log the specific Source IP or API Key used to trigger a "System Reset," leaving no trail of who issued the command.	Medium	Low	Medium
Info Disclosure	API documentation (like Swagger/OpenAPI) is left public, revealing hidden endpoints that leak system metadata.	High	Medium	High
Denial of Service	An attacker performs an "API Stress Attack," exhausting the cloud's compute resources and delaying sensor updates.	High	Medium	High
Elevation of Privilege	An attacker exploits a broken functional level authorization to access the `/admin/system_wipe` endpoint using a basic user token.	Low	Critical	Critical

Component 4: Remote Controls (Gate/Shutoff)

Threat	Scenario	Likelihood	Impact	Risk
Spoofing	An attacker captures and replays a wireless RF signal to open a restricted security gate without a valid credential.	High	Medium,	High
Tampering	A malicious actor physically opens the control box at the dock and jumps the wires to force a shutoff valve into the "Open" position.	Medium	Medium	High
Repudiation	The local actuator firmware does not log "Manual Override" events, allowing someone to shut off water to a slip without a digital record.	Low	Low	Medium
Info Disclosure	The control unit broadcasts its status (e.g., "Valve Open") over an unencrypted 433MHz frequency, allowing anyone nearby to monitor marina operations.	High	Low	Medium
Denial of Service	A signal jammer is placed near the main gate, preventing the remote server from sending a "Close" command during an emergency.	High	Medium	High
Elevation of Privilege	An attacker exploits a vulnerability in the actuator's Bluetooth maintenance port to bypass the "Locked" state and gain administrative	Low	High	High

	control.			

Section 5: Risk Summary

List your findings by risk level:

Critical Risks:

Automated Response Manipulation

1. Description: Combining spoofed sensor data with a DoS attack on the gateway.
2. Why it's Critical: It creates a "blind spot." Operators see "safe" levels while pumps are jammed, leading to massive flooding and vessel damage.

High Risks:

Hydrologic Grid Takeover

1. Description: Using guest Wi-Fi or mobile apps to pivot into the industrial control network.
2. Why it's High: An attacker gains full admin control over sluice gates and main shut-off valves for the entire marina.

Medium Risks:

Environmental Data Tampering

1. Description: Remote or physical alteration of water quality thresholds.
2. Why it's Medium: Undetected toxic runoff leads to heavy environmental fines and long-term ecological har**m**

Section 6: Recommended Mitigations

For each Critical and high-risk, propose a defense:

Risk	Proposed Mitigation	Implementation Complexity
Fake Data & Jamming	Use a backup wire for sensors so they don't only rely on Wi-Fi.	High — must run new cables.
Fake Data & Jamming	Set up an alarm that goes off if a sensor stops sending updates.	Medium — needs a software alert.
Network Hacking	Put the marina controls on a different network than the guest Wi-Fi.	Low — change router settings.
Network Hacking	Make admins use a code from their phone (MFA) to log in.	Low — turn on a security setting.